The role of population size and density in evolutionary rescue

Rescue Crew

Summary to-date from theory meetings

Setting up the problem:

Classical rescue theory poses that the probability of rescue increases monotonically with population size (Orr and Unckless, 2008), because larger population sizes help buffer the population from decline (ecological effect), are generally associated with higher standing genetic variation (from mutation- selection balance) and higher mutation accumulation (more absolute reproduction). Specifically, the dependence has been known to be negative-exponential

$$P_{\text{res}} = 1 - \exp(-\theta N_0) \tag{1}$$

For instance, in a model with two alleles (mutant and wildtype) where the frequency of mutants p_{sgv} is constant, the wildtype declines exponentially at rate $r < 0$ and establishment probability p_{est} doesn't depend on the population size, we have $\theta = (p_{\text{sgv}} + \frac{\mu}{r})p_{\text{est}}$. This negative exponential dependence on N_0 also holds for a variety of other cases, such as for more than two alleles and multi-step rescue (Martin *et al.*, 2013, equation 3.1), a quantitative trait (Anciaux *et al.*, 2019, equation 4), space (Uecker *et al.*, 2014, equation 17), recombination (Uecker and Hermisson, 2016, equation 10), and diploids with variable amounts of sex and selfing (Uecker, 2017, equation 5-6,8)....

Given this clear dependence and the ease of measuring population size in experiments and the field, it is one of the better-known and useful(?) relationships. Experimental work has corroborated the qualitative dependence that increasing population size increases the probability of rescue (Bell and Gonzalez, 2009). However, quantitative tests are few (but see Martin *et al.*, 2013). (EXPERIMENTAL GROUP info on quantitative tests)

However, θ could depend on population size in a variety of cases. In fact the more general formula for the rescue probability (CITE) is given by

$$P_{\text{res}} = 1 - \exp\left(-N_{\text{sgv}} p_{\text{est}}(0) - \int_0^\infty \mu(t) N_w(t) p_{\text{est}}(t) dt\right) \tag{2}$$

where all the different factors could depend on N_0 in complex ways, leading to the exponential factor having a non-linear dependence on N_0. For instance, if the wildtype fitness is density dependent, then it can affect the mutational input (Orr and Unckless, 2008) while density dependence in mutant fitness can alter its establishment probability (Orr and Unckless, 2008;

Uecker *et al.*, 2014; Wilson *et al.*, 2017). Density dependence could be negative (such as due to competition) or positive (such as due to Allee effects). In addition other parameters, such as the frequency of mutants in standing genetic variation (SGV) could also depend on N_0. For instance if N_0 is also the size of the population before the environment changes, it can also affect the p_{sgv} due to drift (Tanaka and Wahl, 2022).

It is important to note that the dependence of theta on population (census) size could be via its density i.e., the number of individuals per unit resource (such as area, food etc.). For instance, the fitness dependence (of the mutant or wildtype) of the probability of rescue is most likely due to density rather than census size, i.e. we would expect competition (a source of negative-density dependence) to be unchanged if we increase the number of individuals but also the amount of available resources. On the other hand, the frequency of mutants in the SGV changes due to the the magnitude of drift which depends on the population size before change. Therefore, it is important to distinguish between this dependence on density and population size, especially when proposing experiments.

In this work, (1) we explore the dependence on θ on population density and size with a focus on distinguishing between the two, (2) use simulations to test if and how this dependence can be statistically detected from data, and (3) propose experiments to do test these. By jointly analysing theory and experimental studies, we provide key insights into a key gap in the evolutionary rescue literature.

Results

Here we explore two models where θ is density-dependent, specifically via fitness of the wildtype and mutant to make the following salient points

1. Population size dependence can be detected from rescue probabilities
2. However, smaller or sparser range of population sizes makes it harder to detect (especially for positive-density dependence in the model and parameter ranges tested here)
3. Despite density dependence in θ (via fitness), the probability of rescue might have the canonical negative exponential dependence on N_0 if the density is kept constant, i.e. $\frac{N_0}{K}$ is constant.

Negative density dependence

We explore a continuous-time model where birth rates of mutants and wildtypes have negative-density dependence

$$\lambda_i(N) \equiv \max\left(b_i \times (1 - \frac{N_m + N_w}{K}), 0\right) \;\; ; \;\; i = w, m$$

while death rates are constant $\mu_i = d_i; i = w, m$. [PD NOTE: The density-dependence here is a bit weird where we assume a common parameter K for the two types but this means very little. I now have a better system with better interpretation of the density regulations as well as more general form of birth and death rates. See Appendix A.3] We denote by $r_i = b_i - d_i$

the mean growth rate in the absence of density dependence. An important thing to note is that the "carrying capacity" (defined as the non-zero equilibrium of the deterministic model) for each type is different, $K_i = K\frac{r_i}{b_i}$.

We start with a population that has no standing genetic variation with N_0 wildtypes. Mutants are generated via mutations at rate μ (per individual, per unit time) and a mutant generated at time t establishes with probability $p_{\text{est}}(t)$. The rescue probability can be computed by considering a deterministic decline of the wildtype and computing P_{res} as in Uecker and Hermisson (2011). We provide the details for $p_{\text{est}}(t)$ in the appendix. Here we show the final form,

$$P_{\text{res}} = 1 - \exp(-\mu \int_0^\infty N_w(t) p_{\text{est}}(t)) \tag{3}$$

$$= 1 - \exp(-\mu \frac{K}{b_w} \int_0^{\frac{n_0}{n_0 - \frac{r_w}{b_w}}} \frac{p_{\text{est}}(x)}{1-x} dx), \tag{4}$$

where $n_0 = \frac{N_0}{K}$. The second equality was obtained with a change of variable from time t to density-related variable $x = \frac{N_w(t)}{N_w(t) - K_w} = \frac{n_0 e^{r_w t}}{n_0 - \frac{r_w}{b_w}}$ [PD NOTE: There might be a more intuitive choice for x that retains the key points but need to think more]. Note that, for a constant K, since $p_{\text{est}}(x) \to 0$ as $x \to 1$, the rescue probability need not go to 1 as N_0 (and consequently n_0) goes to infinity. Further, let $P_\infty = \lim_{N_0 \to \infty} P_{\text{res}} = 1 - \exp(-\mu \frac{K}{b_w} \int_0^1 \frac{p_{\text{est}}(x)}{1-x} dx)$ be the saturating value of the rescue probability. If $P_\infty < 1$ (not sure if this second assumption is required), then we can approximate the rescue probability as

$$P_{\text{res}} = 1 - \exp(- \underbrace{\frac{\theta_1}{\theta_2 + N_0}}_{\theta(N_0) = \tilde{\theta}(n_0)} N_0) \tag{5}$$

where $\theta_1 = -\ln(1 - P_\infty)$ and $\theta_2 = \frac{r_w}{2\mu r_m} \ln(1 - P_\infty)$, which provides an alternate model to fit the data. It is worth noting that both θ_i are proportional to K. So $\theta(N_0)$ is actually only a function of the density n_0, i.e., $\theta(N_0) = \frac{\theta_1/K}{\theta_2/K + n_0} = \tilde{\theta}(n_0)$ such that θ_1/K and θ_2/K don't depend on K or N_0. Therefore, if we change population size while keeping the density constant, we recover the classical negative exponential dependence (Fig. 2). This formulation also allows for the estimation of the strength of competition but not sure if this is useful (see appendix A.2)

We then ran simulations for cases with no competition ($K \to \infty$), and two degrees of competition ($K = 10^4, 3 \times 10^4$). We ran 10^4 replicates for each value of N_0. Each replicate started with N_0 wildtype and was run using a Gillespie algorithm with 5 events: birth and death of each type at rates $b_i(N_w + N_m)N_i$ and $d_i N_i$ respectively and mutation rate from wildtype to mutant at rate μN_w (See github for the code).

We then fit each of these to a negative exponential function (assuming θ is density independent) and to the alternative model. We find that with the measurement of rescue probability across a sufficient range of population, one can detect effects of competition, even with low number of replicates (compare Fig. 1 a and c). However, lower range (Fig. 1 b and e) and sparser data

(Fig. 1 c and f) can make it harder to detect these trends. The alternative model does a much better job at fitting the data. Unfortunately, with noisy data, the support for the alternative model is high even when the true model is the null model with no competition (see Fig. 1 d-e).

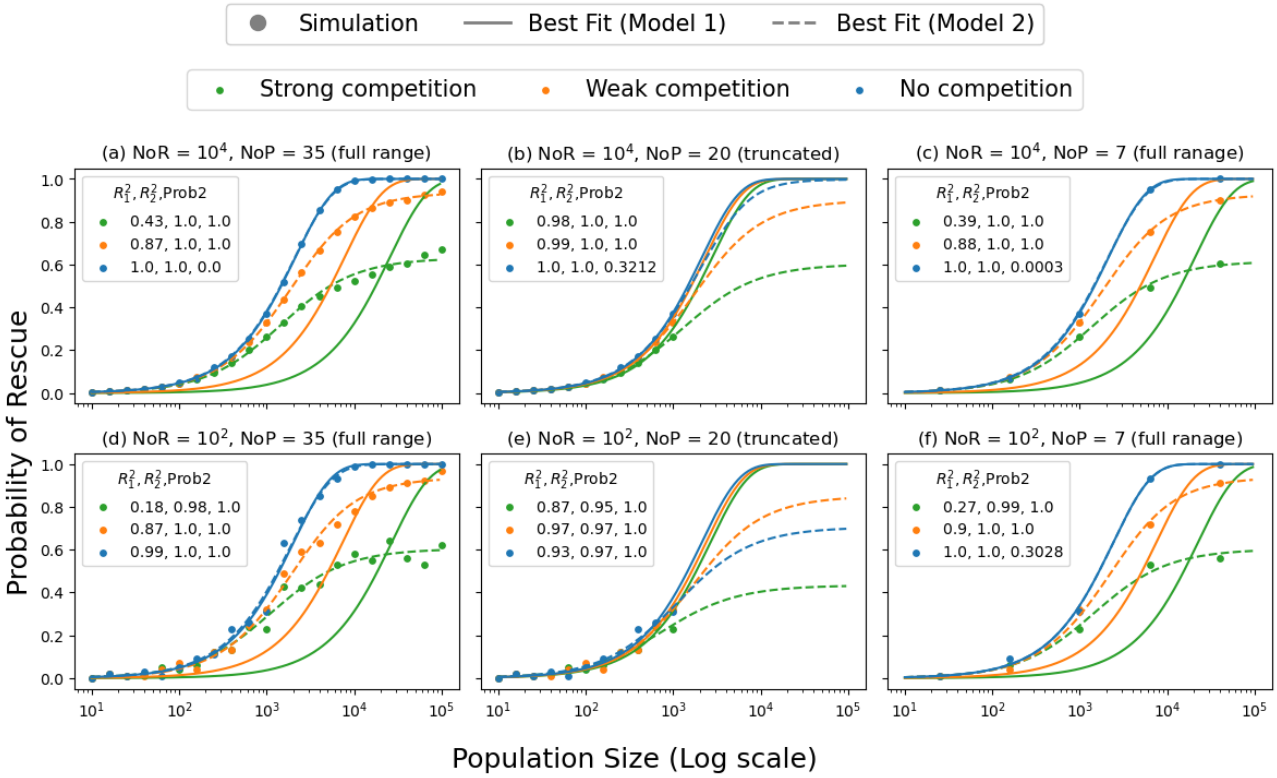

Figure 1: Probability of rescue in a birth-death model with no density dependence ($K = \infty$; blue lines), weak ($K = 3 \times 10^4$; orange lines) and strong negative ($K = 10^4$; green lines) density-dependent birth rates. NoR is the number of replicates and NoP is the number of points for each line. The lines are are the best fit line determined using MLE under a GLM with Binomial distribution and link function $-\log(1-p)$ (as in Martin *et al.* (2013)) with $p = 1 - \exp(-\theta N_0)$ (solid lines) and $p = 1 - \exp(\theta_1(1 + \frac{\theta_2}{N_0})^{-1})$. The goodness of fit measure here is an R_i^2 value computed for each model (there are prob other better options. Martin *et al.* (2013) uses a χ^2 test on something called Deviance but idk what that is atm). Further, Prob2 is the support for model 2 with value between 0 and 1. For computing Prob2, I computed the AIC values for both models and then took the minimum, say $AIC_{\min}$. Then Prob2 $= \frac{\exp((AIC_{\min} - AIC_2)/2)}{\exp((AIC_{\min} - AIC_2)/2) + \exp((AIC_{\min} - AIC_1)/2)}$. Parameters : $b_w = \frac{1+\ln(0.95)}{2}, d_w = \frac{1-\ln(0.95)}{2}, b_m = \frac{1+\ln(1.15)}{2}, d_w = \frac{1-\ln(0.15)}{2}, \mu = 10^{-4}$.

Positive density dependence

Now we consider a model with positive density dependence in fitness. Again, we focus on the case with no SGV so rescue is contingent on *de-novo* mutations. Specifically, we consider a model with mate-finding Allee effect (as in Deraje and Uecker (2025)). This is a discrete time mode with non-overlapping generations with hermaphrodites where the probability of the female gamete being fertilized is given by $1 - \exp(-a(N-1))$ where N is the population size. Here, $a > 0$ quantifies the strength of the Allee effect. So, smaller values of a lead to lower probability of finding mates (strong Allee effect) and $a \to \infty$ is the case with no Allee effect (guaranteed mating). Individuals produce offsprings according to a Poisson distribution with mean $\lambda_i; i =$

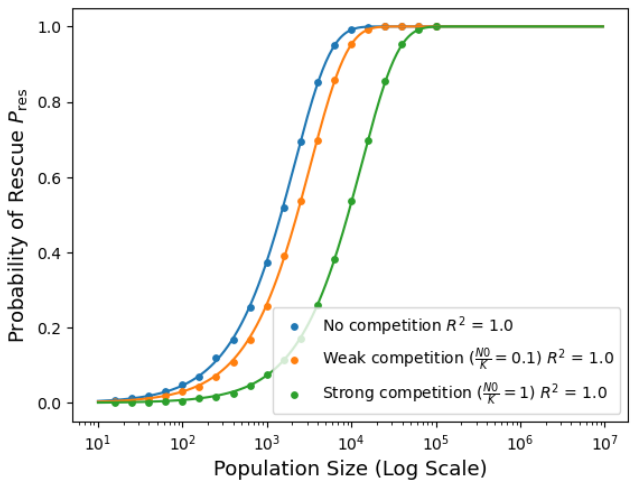

Figure 2: Rescue probability plotted against initial population size, N_0, when carrying capacity is scaled proportional to N_0. We do this for different degree of competition determined by $\frac{N_0}{K}$.

w, m. Offspring inherits the trait of either parent with equal probability. We assume that post enviromental change, the wildtype is subcritical, even if mates are assured, i.e. $\lambda_w < 1$. We provide the model details in the appendix. Since we build the model mechanistically, the "Allee threshold" (defined as the population size below which extinction is guaranteed in a monomorphic population) is different for the two types and more importantly depends on the mean offspring number. Further, for the wildtype it is ∞ (since it is subcritical) and for the mutant it is given by $a_m = -\frac{1}{a}\ln(1 - \frac{1}{\lambda_m})$ (see Eq.8 of Deraje and Uecker (2025))

However, the key point is that it is hard to detect positive density-dependence using just the null model, even when we have high replicates and large range of population sizes (Fig. 3 a and d). However, fitting the model by excluding the saturated part (where $P_{\text{rescue}} = 1$) helps in distinguishing the two much better (Fig. 3 b and e). [PD NOTE: We can probably derive alternative model as in the case of competition but not done yet. Further, $P_{\text{res}} = \dfrac{1 - \exp(-\theta_1 x)}{1 + \exp(-\theta_1(x - x_0))}$ does a decent job (this is a pure guess though)].

Distinguishing between positive and negative density

Note, that the statistics used in the previous sections just detect (when possible) if the rescue probability follows the negative exponential or not. However there are some clear differences between the positive and negative density dependence plots. The main one being that the simulated values the behavior of the difference between the simulated values and the model fit. We plot this in Fig. 4. For positive density dependence (the left figure) the difference goes from negative to positive while for competition (right figure) it is the other way round. Further, for negative-density dependence the saturating values need not be 1 so the difference doesn't go to 0.

Comments and next steps:

1. HU: For intro, it would be weird to not start with the more general equation (Eq. 2) rather than the simpler (Eq. 1).

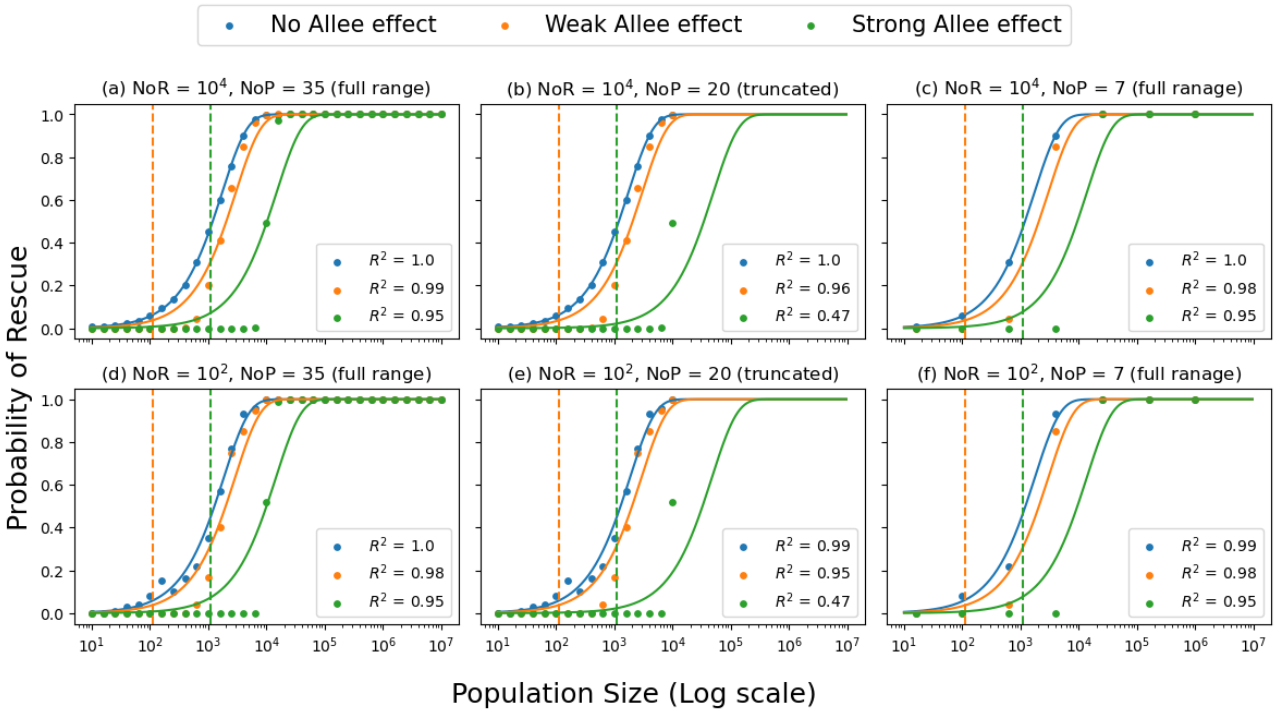

Figure 3: Probability of rescue in a discrete model without density dependence ($a = \infty$; blue lines) and with weak ($a = 0.01$; orange lines) and strong ($a = 0.001$; green lines) mate-finding Allee effect. The dashed line denotes the Allee threshold of a monomorphic mutant population given by $-\frac{1}{a}\ln(1 - \frac{1}{\lambda_m})$. NoR is the number of replicates and NoP is the number of points for each line. The fitted line is an MLE under a GLM with Binomial distribution and link function $-\log(1-p)$ as in Martin *et al.* (2013). The goodness of fit measure here is an R^2 value (there are prob other better options. Martin *et al.* (2013) uses a χ^2 test on something called Deviance but idk what that is atm). Other parameters: $\lambda_w = 0.95, \lambda_m = 1.5, \mu = 10^{-4}$.

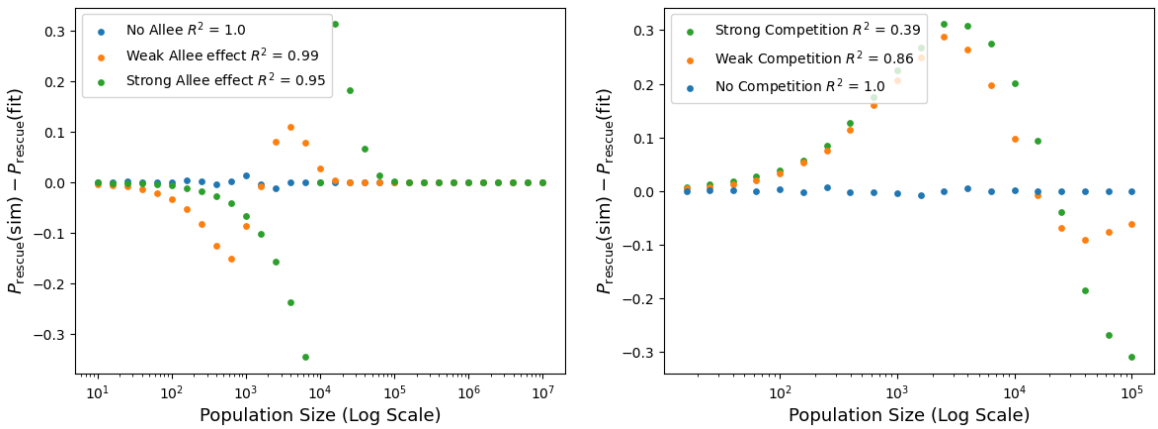

Figure 4: The difference between the rescue probability obtained via simulations. $P_{\text{res}}(\text{sim})$ and the one obtained from fitting the negative exponential model with a constant θ, $P_{\text{res}}(\text{fit})$ for the model with positive (left) and negative (right) density dependence. The parameter values for simulations are given in Figs. 3 and 1, respectively.

PD: Agreed. What's the general starting point in the experimental world? This will determine the phrasing.

2. PD: Consider the statistical techniques a bit more carefully. Given, a key aim is to suggest experiments, it would be useful to spend some time on the statistics of it all. Something along formalizing the patterns of Fig. 4.

PD: I can compute a bunch of things (like the bias etc.) but would like to narrow things down before doing that.

HA: Re: distinguishing between positive/negative DD: If we analyse data using a GLM(M) with a cloglog link function, we can test whether slope significantly differs from 1 (and in what direction) - i.e. if the rescue curve is steeper or shallower than expected under the null model. Members of my group have been doing this in slightly different contexts.

HA: Re: how to test goodness of fit of the null model: I've also used the chi-sq test on deviance before, but it may not be the best option. IIRC there are two problems: (1) deviance increases with the number of data points (number of different pop sizes tested) and so the fit will always appear 'worse' as data is added. (2) If you only test pop sizes at the plateaus where zero or all replicates show rescue, you can get a perfect fit, but it's not very meaningful.

3. LB: Make connection with the "7 types of rarity" paper. Consider how to best frame this, e.g., do we want to show the seven types across a census/density plot and then highlight the 2-3 examples we simulate or allude to which "sections" of the plot might differ from classical expectations, e.g., high density/low population size. Puneeth's current simulations address the first of these questions; I'm not sure about "measuring" the density directly but it's something that I believe we've talked about a touch in our meetings. I think the second question (about metrics rather than processes) would be try to make sure we hit different sections of the "types of rarity", i.e, does density "act" truly independently of population size or is there some interaction there? Four examples could be (density/census): low/small, low/large, high/small, high/large.

   HA: Is this Rabinowitz 1981? I quickly looked it up and saw that it contrasts 3 binary elements. Wide vs. narrow geographic range - I'm not sure whether we can make a direct connection when we aren't dealing with spatial models. I'm also not sure if there's any equivalent of specialist vs. generalist habitat in the models we're dealing with. But maybe I am interpreting the paper too narrowly, I could see there may be a more general point in the (density/census) pairings.

4. HA: We now have two nice case studies simulated by Puneeth. Both deal with density dependent fitness, and no SGV. To discuss: should we add a 3rd (or more) case study with another form of pop size or density dependence? Cf. comment below about the Tanaka and Wahl paper... (IIRC, they deal with a fundamentally different effect where historical pop size shapes the amount of SGV.)

5. PD: We should consider Tanaka and Wahl (2022) paper since it already has a plot of P_{res} as a function of N_0 and the dependence is via size as opposed to density. Potentially, could connect to Laure's experimental work?

   HU: Will think of some other potential options.

   HA: We wondered about making this a 3rd case study - even reproducing their figure (if allowed). Details added below by Puneeth.

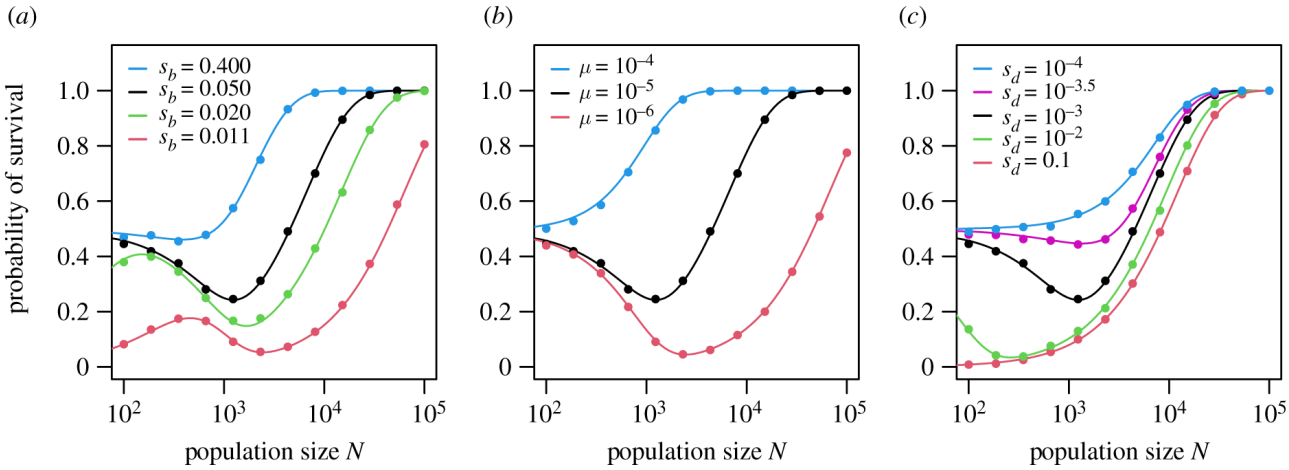

Figure 5: Copied from Tanaka and Wahl!

Eq. 2.10 of Tanaka and Wahl (2022)

$$P_{\text{survival}} = 1 - C(N_0) \exp\left(-N_0 p_{\text{est}} \frac{\mu}{r}\right) \tag{6}$$

where $C(N_0)$ is a ratio of hypergeometric functions and denotes the probability that standing genetic variation is unable to rescue the population. See Fig. 5

- The form is not quite the negative exponential due to the constant $C(N_0)$: The frequency of beneficial mutants in SGV, p_{sgv} depends on N_0 as it is also the pre-change population size. High number of mutants in SGV, given by $p_{\text{sgv}} N_0$ can be due to larger p_{sgv}, as at low population size due to drift or due to larger population size N_0. Non-monotonic dependence on N_0
- Dependence of θ on N_0 :
- Other notes : Sometimes frequency at the time of change can be high such the population is not at risk of extinction (so might not be ER hence the use of survival).

MO: I'm in favour of basically just putting their result in.

6. HA: Are there any other key theory references dealing with density-dependent effects on rescue probability that we should cite?

   (a) For competition: (i) Eq B3 and Fig 5 of Uecker *et al.* (2014), (ii) Eq. 4 and para on "impact of population density" of Wilson *et al.* (2017), (iii) Czuppon *et al.* (2023)

   (b) Marrec and Bitbol (2020) Para on "imapact of population size" (for competition and type of environmental change?)

   (c) PD Fig 3 and 6 of Kanarek and Webb (2010) for spatial Allee effects.

A Birth-death process with competition - Model details

Here we outline the analytical computation of rescue probability for a birth-death process with competition. Assuming the wildtype decays deterministically and $N = N_w + N_m \approx N_w$ we have

$$\begin{aligned}
\frac{dN_w(t)}{dt} &= b_w N_w(1 - \frac{N_w}{K}) - d_w N_w \\
&= (b_w - d_w) N_w (1 - \frac{b_w N_w}{K(b_w - d_w)}) \\
&= r_w N_w (1 - \frac{N_w}{K_w})
\end{aligned}$$

where $K_w = \frac{r_w}{b_w} K$. Then we can solve this logistic growth equation to get

$$N_w(t) = \frac{N_0 e^{r_w t}}{\frac{N_0}{K_w} e^{r_w t} + (1 - \frac{N_0}{K_w})}$$

Now, we can compute the establishment probability, $p_{\text{est}}(t)$ of a mutant that arises at time t by a birth death process. From Eq. of Uecker and Hermisson (2011), we have

$$p_{\text{est}}(t) = 2[1 + \underbrace{\int_t^\infty (\lambda_m + \mu_m)(t') \exp\left(\underbrace{\int_t^{t'} (\lambda_m - \mu_m)(\tau) d\tau}_{I(t,t')}\right) dt'}_{J(t)}]^{-1}$$

Now, we compute $I(t, t')$,

$$\begin{aligned}
I(t, t') &= \int_t^{t'} (\lambda_m - \mu_m)(\tau) d\tau \\
&= \int_t^{t'} (r_m - \frac{b_m}{K} N_w(\tau)) d\tau \\
&= r_m(t' - t) - \frac{b_m N_0}{r_w K} \int_{e^{r_w t}}^{e^{r_w t'}} \frac{dy}{\frac{N_0}{K_w} y + (1 - \frac{N_0}{K_w})} \\
&= r_m(t' - t) - \frac{b_m}{b_w} \ln \left[\frac{\frac{N_0}{K_w} e^{r_w t'} + (1 - \frac{N_0}{K_w})}{\frac{N_0}{K_w} e^{r_w t} + (1 - \frac{N_0}{K_w})} \right] \\
&= r_m(t' - t) - \frac{b_m}{b_w} \ln \left[\frac{1 + \frac{N_0}{K_w - N_0} e^{r_w t'}}{1 + \frac{N_0}{K_w - N_0} e^{r_w t}} \right] \\
\exp(-I(t, t')) &= \exp(-r_m(t' - t)) \left[\frac{1 + \frac{N_0}{K_w - N_0} e^{r_w t'}}{1 + \frac{N_0}{K_w - N_0} e^{r_w t}} \right]^{\frac{b_m}{b_w}}
\end{aligned}$$

We can use this to compute $J(t)$

$$J(t) = \frac{e^{r_m t}}{(1 + A e^{r_w t})^{\frac{b_m}{b_w}}} \int_t^\infty (b_m + d_m - \frac{b_m N_w(t')}{K}) e^{-r_m t'} (1 + A e^{r_w t'})^{\frac{b_m}{b_w}} dt'$$

$$
\begin{aligned}
\frac{(1+Ae^{r_w t})^{\frac{b_m}{b_w}}}{e^{r_m t}} J(t) &= (b_m+d_m)\int_t^\infty e^{-r_m t'}(1+Ae^{r_w t'})^{\frac{b_m}{b_w}}dt' - \frac{b_m A K_w}{K}\int_t^\infty \frac{e^{r_w t'}e^{-r_m t'}}{1+Ae^{r_w t'}}(1+Ae^{r_w t'})^{\frac{b_m}{b_w}}dt' \\
&= (b_m+d_m)\int_t^\infty e^{-r_m t'}(1+Ae^{r_w t'})^{\frac{b_m}{b_w}}dt' - \frac{b_m A K_w}{K}\int_t^\infty e^{(r_w-r_m)t'}(1+Ae^{r_w t'})^{\frac{b_m}{b_w}-1}dt' \\
&= -\frac{(b_m+d_m)}{r_w}\int_0^{e^{r_w t}} u^{-\frac{r_m}{r_w}-1}(1+Au)^{\frac{b_m}{b_w}}du + \frac{b_m A K_w}{r_w K}\int_0^{e^{r_w t}} u^{-\frac{r_w}{r_m}}(1+Au)^{\frac{b_m}{b_w}-1}du
\end{aligned}
$$

where $u = e^{r_w t'}$. These integrals have a solution given by hypergeometric function (see Eq.3.194.1 in Gradshteyn and Ryzhik (2014))

$$
\int_0^y u^{p-1}(1+Au)^{-q} = \frac{y^p}{p}\underbrace{{}_2F_1(q,p,1+p,-Ay)}_{F_{p,q}(-Ay)}.
$$

Therefore, we have

$$
\frac{(1+Ae^{r_w t})^{\frac{b_m}{b_w}}}{e^{r_m t}} J(t) = -\frac{(b_m+d_m)}{r_w}\frac{e^{-r_w t\frac{r_m}{r_w}}}{-r_m/r_w}F_{-\frac{r_m}{r_w},-\frac{b_m}{b_w}}(-Ae^{r_w t}) + \frac{b_m}{b_w}A\frac{e^{r_w t(1-\frac{r_m}{r_w})}}{1-\frac{r_m}{r_w}}F_{1-\frac{r_m}{r_w},1-\frac{b_m}{b_w}}(-Ae^{r_w t})
$$

$$
J(t) = \frac{1}{(1+Ae^{r_w t})^{\frac{b_m}{b_w}}}\left[\underbrace{\frac{(b_m+d_m)}{r_m}F_{-\frac{r_m}{r_w},-\frac{b_m}{b_w}}(-Ae^{r_w t}) + \frac{b_m}{b_w}A\frac{e^{r_w t}}{1-\frac{r_m}{r_w}}F_{1-\frac{r_m}{r_w},1-\frac{b_m}{b_w}}(-Ae^{r_w t})}_{H(-Ae^{r_w t})}\right]
$$

To compute the rescue probability is we make the change of variable, $x(t) = -Ae^{r_w t} = \frac{N_0 e^{r_w t}}{N_0-K_w}$, which gives $dx = \frac{x}{r_w}dt$. Further $J(t)$ only depends on t via $x(t)$ so we can write it as J(x)

$$
\begin{aligned}
P_{\text{res}} &= 1-\exp\left(-\mu\int_0^\infty N_w(t)p_{\text{est}}(t)dt\right) \\
&= 1-\exp\left(-\mu\int_{\frac{N_0}{N_0-K_w}}^0 \frac{K_w}{1-\frac{1}{x}}\frac{2}{1+J(x)}\frac{dx}{r_w x}\right) \\
&= 1-\exp\left(-\mu\frac{K_w}{r_w}\int_0^{\frac{N_0}{N_0-K_w}} \frac{1}{1-x}\frac{2}{1+J(x)}dx\right) \\
&= 1-\exp\left(-\mu\frac{K}{b_w}\int_0^{\frac{n_0}{n_0-\frac{r_w}{b_w}}} \frac{1}{1-x}\frac{2}{1+J(x)}dx\right)
\end{aligned}
$$

where $n_0 = \frac{N_0}{K}$.

While this does have an exact closed form solution. It can be approximated quite well as below

$$
P_{\text{res}} = 1-\exp(-\frac{\theta_1}{N_0+\theta_2}N_0)
$$

where $\theta_1 = -\ln(1-P_\infty)$ and $\theta_2 = -\frac{r_w}{2\mu r_m}\theta_1$, where $P_\infty = \lim_{N_0\to\infty} P_{\text{res}}(N_0)$. We obtained this by analysing the behaviour of $p_{\text{est}}(x)$ near $x=0$ and $x=1$. If we have the exact value of P_∞, this approximation does a great job at approximating the rescue probability (see Fig. 6). However,

while we do have an approximation for P_∞, its crude. However, while estimating from data P_∞ is infered correctly. Below we provide the details of analysis that led to this form.

A.1 Approximating the density integral

The taylor series near $x = 0$ of $p_{\text{est}}(x) = \frac{2}{1+J(x)}$ (obatained via Mathematica) is given by

$$p_{\text{est}}(x) = \underbrace{\frac{r_m}{b_m}}_{C_0} + \underbrace{\frac{r_m}{b_m}\frac{d_m}{b_w}\frac{r_w}{r_m - r_w}}_{C_1} x + \mathcal{O}(x^2)$$

while its behaviour near $x = 1$ is characterised by

$$p_{\text{est}}(x) = 2H^{-1}(1-x)^{\frac{b_m}{b_w}} + \mathcal{O}((1-x)^{\frac{2b_m}{b_w}})$$

where $H \equiv H(1) = \frac{(b_m+d_m)}{r_m} F_{-\frac{r_m}{r_w},-\frac{b_m}{b_w}}(1) - \frac{b_m}{b_w}\frac{1}{1-\frac{r_m}{r_w}} F_{1-\frac{r_m}{r_w},1-\frac{b_m}{b_w}}(1)$ It is important to note that $\frac{b_m}{b_w} > 1$. In fact if the mutant and wildtype have weak selection and we assume $b_i = \frac{1}{2} + \frac{r_i}{2}$ and $d_i = \frac{1}{2} - \frac{r_i}{2}$, we can take $\frac{b_m}{b_w} \approx 1$. Under this approximation, $H \approx \dfrac{(1-r_m)r_w}{(r_w - r_m)r_m}$, $C_0 \approx 2r_m$ and $C_1 \approx \frac{2r_m r_w}{r_m - r_w}$.

We use the two behaviours to approximate $p_{\text{est}}(x)$ as

$$p_{\text{est}}(x) = \begin{cases} C_0 + C_1 x & \text{if x} \leq x_{\text{cut}} \\ 2H^{-1}(1-x) & \text{if x} > x_{\text{cut}} \end{cases}$$

where x_{cut} is given by

$$\begin{aligned} C_0 + C_1 x_{\text{cut}} &= 2H^{-1}(1-x_{\text{cut}})^{\frac{b_m}{b_w}} \\ x_{\text{cut}} &\approx \frac{2H^{-1} - C_0}{2H^{-1} + C_1} \\ &= 1 + \frac{r_w(1-r_m)}{r_m - 2r_w + r_w^2} \end{aligned}$$

Now, define $x_0 := \frac{n_0}{n_0 - \frac{r_w}{b_w}}$. Then

$$\begin{aligned} P_{\text{res}}(N_0) &= 1 - \exp(-\mu \frac{K}{b_w} \int_0^{x_0} \frac{2dx}{(1-x)(1+J(x))}) \\ &= 1 - \exp(-\mu \frac{K}{b_w} \left[\int_0^{\min(x_0, x_{\text{cut}})} \frac{C_0 + C_1 x}{1-x} dx + \int_{\min(x_0, x_{\text{cut}})}^{x_0} 2H^{-1}(1-x)^{\frac{b_m}{b_w}-1} dx \right]) \\ &= 1 - \exp(-\mu \frac{K}{b_w} \left[-C_1 x_{\min} - (C_0 + C_1)\ln(1 - x_{\min}) + \frac{2H^{-1}}{b_m/b_w}((1-x_{\min})^{\frac{b_m}{b_w}} - (1-x_0)^{\frac{b_m}{b_w}}) \right]) \end{aligned}$$

where we used integration by partial fractions for the first integral and the second integral is of the form $u^m du$. This in turn gives

$$P_\infty = \lim_{N_0 \to \infty} P_{\text{res}}(N_0) \approx 1 - \exp(-\mu \frac{K}{b_w} \left[-C_1 x_{\text{cut}} - (C_0 + C_1)\ln(1 - x_{\text{cut}}) + \frac{2H^{-1}}{b_m/b_w}((1-x_{\text{cut}})^{\frac{b_m}{b_w}}) \right])$$

Now note that when $x_0 = \frac{n_0}{n_0 - \frac{r_w}{b_w}} << 1$, then

$$\begin{aligned}
P_{\text{res}} &\approx 1 - \exp(-\mu \frac{K_w}{r_w}[-C_1 x_0 - (C_0 + C_1)\ln(1 - x_0)]) \\
&\approx 1 - \exp(-\mu \frac{K_w}{r_w}[-C_1 x_0 + (C_0 + C_1) x_0)]) \\
&= 1 - \exp(-2\mu \frac{K}{b_w} r_m x_0) \\
&= 1 - \exp(-2\mu \frac{K}{b_w} r_m \frac{n_0}{n_0 - \frac{r_w}{b_w}}) \\
&\approx 1 - \exp(-2\mu \frac{1}{b_w} r_m \frac{N_0}{-\frac{r_w}{b_w}}) \\
&= 1 - \exp(2\mu \frac{r_m}{r_w} N_0)
\end{aligned}$$

Therefore, the function grows at a rate $2\mu \frac{r_m}{r_w}$ early on, while when $x_0 \approx 1$, the behaviours is given by

$$P_{\text{res}} = 1 - \exp(\ln(1 - P_\infty) + \frac{2\mu K}{b_m H}(1 - x_0)) \tag{7}$$

Note, that once $x_0 \approx 1$, the dominating behaviour is the saturating value (the growth is very slow of the order of $\frac{1}{N_0}$). So the two features to capture are the early growth and the saturating value.

$$P_{\text{res}}(N_0) = 1 - \exp(-\theta_1 \frac{N_0}{N_0 + \theta_2}) \tag{8}$$

Equating the early growth and the limiting value of this function with those obtained from the approximations, we get $\theta_1 = -\ln(1 - P_\infty)$ and $\dfrac{\theta_1}{\theta_2} = -2\mu \dfrac{r_m}{r_w}$.

A.2 Inferring K

Note that potentially, one could infer K from θ_1 and θ_2 if we know atleast 2 of these; μ, r_w or r_m. To see this note

$\theta_1 \approx 2\mu K f(r_w, r_m)$ where $f(r_w, r_w) = \displaystyle\int_0^1 \frac{2}{1 - x + H_{approx}(r_m, r_w, x)} dx$ with $H_{\text{approx}}(r_m, r_w, x) = \frac{1}{r_m} F_{-\frac{r_m}{r_w}, -1}(x) - \frac{x}{1 - \frac{r_m}{r_w}} F_{1 - \frac{r_m}{r_w}, 0}(x)$

Note that MLE estimation of $\theta = (\theta_1, \theta_2)$ doesn't always converge well, so we start with different starting points for the minimization until we get the best R^2 value. This does a great job at estimating r_m and K (see Fig. 7)

A.3 Generalizing the density regulation parameterization

The current version is a bit weird where K is common to both wildtype and mutant but they have different density dependence in mean growth rate. To see this more clearly note that at the moment the density dependent growth rate

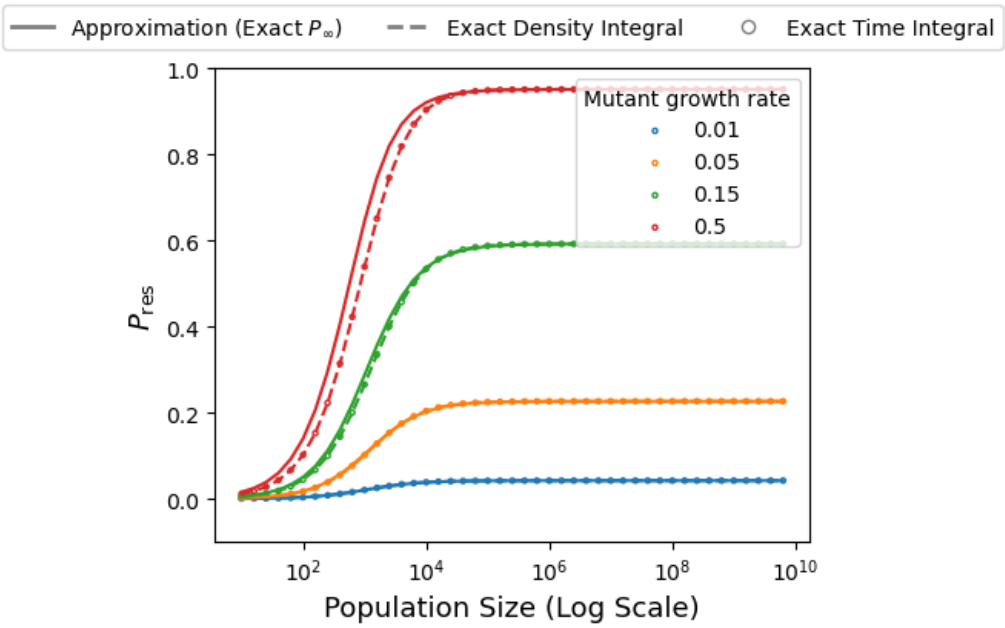

Figure 6: Analytical approximation of rescue probability. The solid line is the approximation given by Eq. 8 while the empty circles and dashed lines are exact numerical integration of the time and density integral respectively for different mutant growth rates s_m (different colors) with $b_m = \frac{1}{2}+\frac{1}{2}\ln(1+s_m)$ and $d_m = \frac{1}{2}-\frac{1}{2}\ln(1+s_m)$ (Note $s_m \approx r_m$). Parameters: $b_w = \frac{1}{2}+\frac{1}{2}\ln(0.95)$, $d_w = \frac{1}{2}-\frac{1}{2}\ln(0.95)$, $\mu = 10^{-4}$, $K = 10^4$.

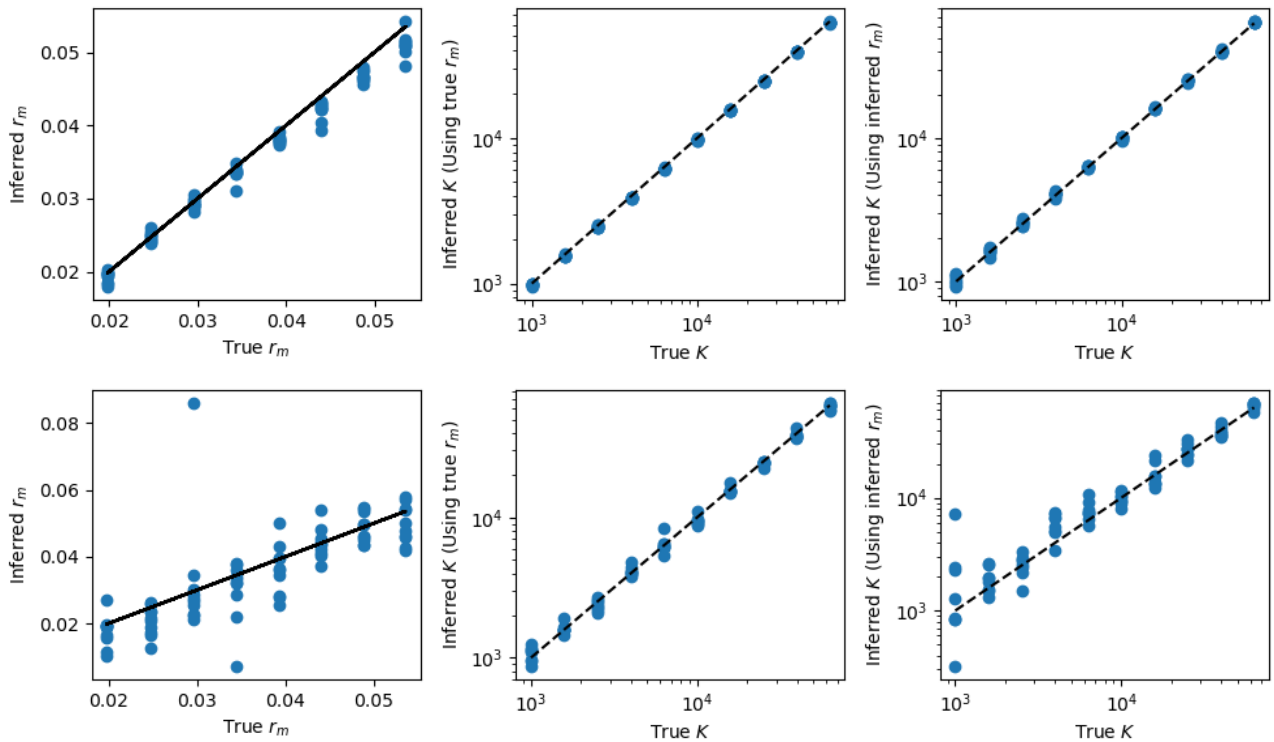

Figure 7: Inferring K: Top row is with 10^4 replicates while the bottom row is for 10^2 replicates

$$\lambda_i - \mu_i = r_i - \frac{b_i}{K} N_{\text{total}}$$

Therefore, the dependence of mean per-capita growth rate is different for the mutant and wildtype (i.e. the coefficient of N_{total} is different). This kind of renders K a "confusing" parameter. A better description would allow K to be different for the wildtype and mutant.

To avoid confusion with previous notation we consider the following parametrization that allows different

$$\lambda_i - \mu_i = r_i - \frac{N_{\text{tot}}}{\kappa_i}$$

Now if $\gamma_w = \gamma_m$, then density regulation is identical between the two types. Further, for full generality we allow the split between birth and rates to be general

$$\lambda_i = b_i - (1-\alpha_i)\frac{N_{\text{tot}}}{\kappa_i} \quad ; \quad \mu_i = d_i + \alpha_i \frac{N_{\text{tot}}}{\kappa_i}$$

Then we get,

$$P_{\text{res}} = 1 - \exp\left(\mu\kappa_w \int_0^{\frac{n_0}{n_0 - r_w}} \frac{2}{(1-x)(1+J(x))} dx\right) \tag{9}$$

$$\approx 1 - \exp(-\frac{\theta_1}{N_0 + \theta_2} N_0) \tag{10}$$

$$\tag{11}$$

where $n_0 = \frac{N_0}{\kappa_w}$, $J(x) = (1-x)^{-\frac{\kappa_w}{\kappa_m}} \left[\frac{b_m + d_m}{r_m} F_{-\frac{r_m}{r_w}, -\frac{\kappa_w}{\kappa_m}}(x) + \frac{\kappa_w}{\kappa_m}(2\alpha_m - 1)\frac{x}{1 - \frac{r_m}{r_w}} F_{1-\frac{r_m}{r_w}, 1-\frac{\kappa_w}{\kappa_m}}(x)\right]$, $\theta_1 = -\ln(1 - P_\infty) = \mu\kappa_w \int_0^1 \frac{2}{(1-x)(1+J(x))} dx$ and $\frac{\theta_1}{\theta_2} = \mu \frac{1}{b_m} \frac{r_m}{r_w}$ which is identical to the previous expression except for $J(x)$.

B Mate-finding Allee Effect - Model details

Consdier a hermaphroditic population that is subject to a mate-finding Allee effect and competition. Let n_t^w and n_t^m be the number of wildtype and mutant individuals at generation t. Assuming discrete non-overlapping generations, the following steps occur

1. Finding a mate: $n_t^x; x = w, m$ are the number of type x individuals that mated

$$n_{t,\text{mated}}^x \sim \text{Bin}(n_t^x, 1 - \exp(-a n_t)) \ ; \ x = w, m$$

2. Type of mate: n_t^{xy} is the number of mate pairs where x provides the female gamete and y provides the male gamete.

$$n_t^{xy} \sim \text{Bin}(n_t^x, \frac{n_t^y}{n_t})$$

3. The number of offsprings pre-mutation is given by

$$n_{t+1,\text{pre-mut}}^w \sim \text{Poi}\left((n_t^{ww} + \frac{1}{2} n_t^{wm})\lambda_w + \frac{1}{2} n_t^{mw} \lambda_m\right)$$
$$n_{t+1,\text{pre-mut}}^w \sim \text{Poi}\left((n_t^{mm} + \frac{1}{2} n_t^{mw})\lambda_m + \frac{1}{2} n_t^{wm} \lambda_w\right)$$

4. After mutations

$$
\begin{aligned}
n^w_{t+1} &\sim \mathrm{Bin}(n^w_{t+1,\text{pre-mut}}, 1-\mu) + \mathrm{Bin}(n^w_{t+1,\text{pre-mut}}, \mu) \\
n^m_{t+1} &= n_{t+1,\text{pre-mut}} - n^w_{t+1}
\end{aligned}
$$

References

Anciaux Y, Lambert A, Ronce O, Roques L, Martin G. 2019. Population persistence under high mutation rate: from evolutionary rescue to lethal mutagenesis. Evolution. 73:1517–1532.

Bell G, Gonzalez A. 2009. Evolutionary rescue can prevent extinction following environmental change. Ecology letters. 12:942–948.

Czuppon P, Day T, Débarre F, Blanquart F. 2023. A stochastic analysis of the interplay between antibiotic dose, mode of action, and bacterial competition in the evolution of antibiotic resistance. PLoS Computational Biology. 19:e1011364.

Deraje P, Uecker H. 2025. How do mate-finding allee effects affect evolutionary rescue? bioRxiv. pp. 2025–09.

Gradshteyn IS, Ryzhik IM. 2014. *Table of integrals, series, and products*. Academic press.

Kanarek AR, Webb CT. 2010. Allee effects, adaptive evolution, and invasion success. Evolutionary Applications. 3:122–135.

Marrec L, Bitbol AF. 2020. Adapt or perish: Evolutionary rescue in a gradually deteriorating environment. Genetics. 216:573–583.

Martin G, Aguilée R, Ramsayer J, Kaltz O, Ronce O. 2013. The probability of evolutionary rescue: towards a quantitative comparison between theory and evolution experiments. Philosophical Transactions of the Royal Society B: Biological Sciences. 368:20120088.

Orr HA, Unckless RL. 2008. Population extinction and the genetics of adaptation. The American Naturalist. 172:160–169.

Tanaka MM, Wahl LM. 2022. Surviving environmental change: when increasing population size can increase extinction risk. Proceedings of the Royal Society B. 289:20220439.

Uecker H. 2017. Evolutionary rescue in randomly mating, selfing, and clonal populations. Evolution. 71:845–858.

Uecker H, Hermisson J. 2011. On the fixation process of a beneficial mutation in a variable environment. Genetics. 188:915–930.

Uecker H, Hermisson J. 2016. The role of recombination in evolutionary rescue. Genetics. 202:721–732.

Uecker H, Otto SP, Hermisson J. 2014. Evolutionary rescue in structured populations. The American Naturalist. 183:E17–E35.

Wilson BA, Pennings PS, Petrov DA. 2017. Soft selective sweeps in evolutionary rescue. Genetics. 205:1573–1586.